

TRISTAN GABL

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EDUCATION

Computational Science and Engineering at ETH Zürich is a comprehensive interdisciplinary program combining applied mathematics, computer science, and engineering, that can be focused on robotics, machine learning, and numerical methods.

ETH Zürich Zurich, Switzerland	2025 — present
Master of Science ETH in Computational Science and Engineering	GPA 5.54/6 (current)
Robot Dynamics	Robot Learning
Probabilistic Artificial Intelligence	Advanced Numerical Methods
Recursive Estimation	Models of Motion
ETH Zürich Zurich, Switzerland	2021 — 2025
Bachelor of Science ETH in Computational Science and Engineering	GPA 5.01/6
Computer Vision	3D Vision
International School Kufstein Tirol, Austria	2016 — 2020
Reifeprüfung mit ausgezeichnetem Erfolg & International Baccalaureate Graduated with dual diplomas from a bilingual school in Tirol, Austria.	
German International School of Silicon Valley California, USA	2017 — 2018
Year abroad, 10th grade	

WORK EXPERIENCE

Additionally two years of hands-on engineering experience at one of the leading Formula Student teams — see Student Organizations below.

Research Assistant at Computer Vision and Geometry Group, ETH Zürich Zurich, Switzerland	2026
Collecting and pre-processing camera and LiDAR data for a computer vision research project.	
Teaching Assistant at ETH Zürich Zurich, Switzerland	2023 — present
Teaching assistant for Computer Science 1 & 2 across several semesters.	
Paramedic at Red Cross at Civil Service Austria	2020 — 2021
Gained experience in taking responsibility in critical situations.	

STUDENT ORGANIZATIONS

Module Lead for Perception, AMZ Racing, Driverless Zurich, Switzerland	2025 — 2026
- Lead the next generation of team members responsible for all perception related subsystems. - Planned and led weekly meetings for timeline management and to ensure knowledge transfer. - Ensured coherence of development goals across the perception, estimation, and controls modules.	
Core Member for Sensor Fusion, Perception, AMZ Racing, Driverless Zurich, Switzerland	2024 — 2025
- Developed bird's eye view sensor fusion model, combining LiDAR, camera, and odometry data for landmark detection at high-speed driving (20 m/s) within latency budget. - Improved and maintained performance of baseline detection methods (LiDAR only, camera only) across incoming software and hardware changes of other modules and the car.	

- Focused on good coding practices to simplify debugging, runtime improvements, and knowledge transfer.

PROJECTS

Cloth folding using a diffusion policy on a SO-101 (github)

Robot Learning

- Established a data collection workflow for imitation learning, used to train and verify diffusion policies.
- Refactored the codebase for use of an unsupported vision backbone in combination with a lighter flow matching model, for effective deployment without a dedicated GPU.
- Generalised model across towel colours using data augmentation leveraging SAM 2.

CuSSS: Cuda implementation of SSS variants (github)

Large-Scale AI Engineering — Awarded a letter of distinction

- Implemented native Cuda kernels for variants of SSS activation functions, focused on runtime.
- Validated kernels against PyTorch autograd equivalents and verified speedups using benchmarks and Nsight Systems profiling.

Action recognition using 3D hand-object contact maps (github)

3D Vision

- Predicted performed actions from short sequences of hand and object vertices on the egocentric H2O dataset.
- Compared two-stage and end-to-end training of MLP and transformer models, using thresholded contact and distance maps as an intermediate representation.

Software pipeline for 3D scene graphs (github)

Bachelor's thesis

- Developed code for combining point clouds, 2D panoptic segmentation, and MLP output to produce 3D scene graphs.

SKILLS

Languages: English • German

Technologies: C • C++ • Python • PyTorch • ROS2 • Cuda • Julia • MATLAB • JavaScript • Git • Linux • Final Cut Pro

Interests: Robot learning • Autonomous systems • Hardware involved projects